Chapter 3 (Part 2)
Number Theory and Cryptography

MAD101

Ly Anh Duong

duongla3@fe.edu.vn

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Definition 1.1

If a and b are integers with $a \neq 0$, we say that a divides b if there is an integer c such that $b = ac$ (or equivalently, if $\frac{b}{a}$ is an integer).

1. a is a factor or divisor of b.
2. b is a multiple of a.
3. **Notation:** $a \mid b$: a divides b, $a \nmid b$: a does not divide b.

Example 1.1. $3 \nmid 13$, $4 \mid 12$

Example 1.2. Let n and d be positive integer. How many positive integers not exceeding ($\leq$) n are divisible by d?

Solution. Suppose $d|m \implies m = kd$. We have $0 < kd \leq n \implies k \leq \frac{n}{d}$.

The number of those integers are $\lfloor \frac{n}{d} \rfloor$

Theorem 1.1

Let a, b, and c be integers, where $a \neq 0$. Then

i. if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

ii. if $a \mid b$, then $a \mid bc$ for all integers c;

iii. if $a \mid b$ and $b \mid c$, then $a \mid c$.

Corollary 1.0

If a, b, and c are integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

Theorem 1.2

Let a be an integer and d a positive integer. Then there are unique integers q and r such that

$$\boxed{a = dq + r, 0 \leq r < d}$$

- d: **divisor** (số chia),
- a: **dividend** (số bị chia),
- $\underbrace{q}_{=a\,div\,d}$: **quotient** (thương),
- $\underbrace{r}_{=a\,mod\,d}$: **remainder** (số dư).

Example 1.3.

1. Find the quotient and the remainder in the division of 101 by 11.
2. Find the quotient and the remainder in the division of –11 by 3.

Q1. What are $-17\,div\,5$ and $-17\,mod\,5$?

a. -3 and 2
b. -4 and 3
c. -3 and -2
d. 3 and 2

Q2. How many integers in $\{1, 2, 3, \ldots, 100\}$ are divisible by 2 but not by 5?

a. 39
b. 51
c. 49
d. 40

The Congruences (đồng dư)

Definition 1.2

If a and b are integers and m is a positive integer, then a is congruent to b modulo m if

$$m \mid ab$$

Notation.

- $a \equiv b \pmod{m}$: a is congruent to b modulo m.
- $a \not\equiv b \pmod{m}$: a and b are not congruent modulo m.

Example 1.4. Are the following congruences true?

a. $17 \equiv 5 \pmod{6}$

b. $24 \equiv 14 \pmod{6}$

Theorem 1.3

Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m} \iff a \bmod m = b \bmod m$.

Theorem 1.4

Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m} \iff \exists k \in \mathbb{Z} : a = b + km$.

Theorem 1.5

Let m is a positive integer. If $\begin{cases} a \equiv b \pmod{m} \\ c \equiv d \pmod{m} \end{cases}$ then $\begin{cases} a + c \equiv b + d \pmod{m} \\ ac \equiv bd \pmod{m} \end{cases}$

Corollary 1.0

Let a and b be integers, and let m be a positive integer. Then

1. $(a + b) \pmod{m} = (a \text{ mod } m + b \text{ mod } m) (\text{mod } m)$
2. $(ab) \pmod{m} = ((a \text{ mod } m)(b \text{ mod } m)) (\text{mod } m)$

A	**B**	**C**	**D**	**E**	**F**	**G**	**H**	**I**	**J**
0	1	2	3	4	5	6	7	8	9
K	**L**	**M**	**N**	**O**	**P**	**Q**	**R**	**S**	**T**
10	11	12	13	14	15	16	17	18	19
U	**V**	**W**	**X**	**Y**	**Z**				
20	21	22	23	24	25				

$$f:\mathbb{Z}_{26}\to\mathbb{Z}_{26}, f(p)=(p+k)\bmod 26\text{: } \textbf{bijection}$$

- **Encryption** (mã hóa) $f(p)=(p+k)\bmod 26$.
- **Decryption** (giải mã) $f^{-1}(p)=(p-k)\bmod 26$.

Example 1.5. What is the secret message produced from the message "MEET YOU IN THE PARK" using the Caesar cipher (k=3)?

Example 1.6. Decrypt the ciphertext message "LEWLYPLUJL PZ H NYLHA ALHJOLY" that was encrypted with the shift cipher with shift k = 7.

Applications of Congruences: Pseudorandom Numbers

$$\boxed{x_{n+1} = (ax_n + c) \bmod m}$$

- a: **multiplier** (nhân tử) $(2 \leq a < m)$
- c: **increment** (số gia) $(0 \leq c < m)$
- x_0: **seed** (hạt giống) $(0 \leq x_0 < m)$
- m: **modulus** (mođun)

Example 1.7. (**Random numbers from 0 to 8**)
Find the sequence of pseudorandom numbers generated by the linear congruential method with modulus m = 9, multiplier a = 7, increment c = 4, and seed $x_0 = 3$.

Solution. $m = 9, a = 7, c = 4, x_0 = 3 \implies x_{n+1} = 7x_n + 4 \implies 3, 7, 8, 6, 1, 2, \ldots$

Applications of Congruences: Hashing Function

$$\boxed{H(k) = k \bmod m}$$

Using in searching data in memory

- k: data searched,
- m: memory block.

Example 1.8. Find the memory locations assigned by the hashing function $h(k) = k \bmod 111$ to the records of customers with Social Security numbers 064212848 and 037149212.

Solution.

$h(064212848) = 064212848 \bmod 111 = 14 \rightarrow$memory location 14.

$h(037149212) = 037149212 \bmod 111 = 65 \rightarrow$ memory location 65.

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Theorem 2.1

Let $b > 1$, $(b \in \mathbb{Z})$. Then if n is a positive integer, it can be expressed uniquely in the form

$$\boxed{n = a_k b^k + a_{k-1} b^{k-1} + \ldots + a_1 b + a_0.\ \text{Denote } (a_k a_{k-1} \ldots a_1 a_0)_b}$$

where k is a nonnegative integer, $a_0, a_1, \ldots, a_k$ are nonnegative integers less than b, and $a_k \neq 0$.

Example 2.1.

a. $(241)_{10} = ? \rightarrow (241)_{10} = 2.10^2 + 4.10^1 + 1.10^0 = 241$

b. $(7016)_8 = ? \rightarrow (7016)_8 = 7.8^3 + 0.8^2 + 1.8^1 + 6.8^0 = 3598 = (3598)_{10}$

c. Suppose that the **letters A through F represent the digits corresponding to the numbers 10 through 15** (in decimal notation). Find $(2AE0B)_{16} \rightarrow = 175627$

Common Bases Expansions

TABLE 1 Hexadecimal, Octal, and Binary Representation of the Integers 0 through 15.

Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
Octal	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
Binary	0	1	10	11	100	101	110	111	1000	1001	1010	1011	1100	1101	1110	1111

Example 2.2.

a. Find 2 expansion of $(241)_{10}$

b. Find 8 expansion of $(12345)_{10}$

c. Find 16 expansion of $(177130)_{10}$

Constructing Base b Expansions

ALGORITHM 1 Constructing Base b Expansions.

procedure *base b expansion*(n, b: positive integers with $b > 1$)
$q := n$
$k := 0$
while $q \neq 0$
  $a_k := q$ **mod** b
  $q := q$ **div** b
  $k := k + 1$
return $(a_{k-1}, \ldots, a_1, a_0)$ $\{(a_{k-1} \ldots a_1 a_0)_b$ is the base b expansion of $n\}$

Example 2.3. Find base 2 expansion of 241.

Solution of Example 2.3

$q = 241 \neq 0$

k=0: $a_0 = 241 \bmod 2 = 1, q = 241 \text{ div } 2 = 120 \neq 0$

k=1: $a_1 = 0, q = 60$

k=2: $a_2 = 0, q = 30$

k=3: $a_3 = 0, q = 15$

k=4: $a_4 = 1, q = 7$

k=5: $a_5 = 1, q = 3$

k=6: $a_6 = 1, 1 = 1$

k=7: $a_7 = 1, q = 0$

$\Longrightarrow 241 = (11110001)_2$

Quiz

2 Integer Representations and Algorithms

Q1. Find base 7 expansion of 186.

a. 354

b. 331

c. 413

d. None of these

Q2. Find the binary format of $(1010)_3$.

a. 11110

b. 11111

c. 100000

d. None of these

Remark 2.1.

$$\begin{cases} 0+0=0 \\ 1+0=1 \\ 1+1=0\,(\text{remind } 1) \end{cases}$$

Example 2.4.

Add $a=(1110)_2$ and $b=(1011)_2$

```
 1 1 1 0
   1 1 1 0 (a)
 + 1 0 1 1 (b)
 1 1 0 0 1 (s)
```

ALGORITHM 2 Addition of Integers.

procedure *add*(a, b: positive integers)
{the binary expansions of a and b are $(a_{n-1}a_{n-2}\ldots a_1a_0)_2$
and $(b_{n-1}b_{n-2}\ldots b_1b_0)_2$, respectively}
$c := 0$
for $j := 0$ **to** $n-1$
$\quad d := \lfloor (a_j + b_j + c)/2 \rfloor$
$\quad s_j := a_j + b_j + c - 2d$
$\quad c := d$
$s_n := c$
return $(s_0, s_1, \ldots, s_n)$ {the binary expansion of the sum is $(s_ns_{n-1}\ldots s_0)_2$}

Solution of Example 2.4

$$c = 0$$

$$j = 0 : d = \lfloor \frac{0+1+0}{2} \rfloor = \lfloor 0.5 \rfloor = 0,\ s_0 = 1, c = 0$$

$$j = 1 : d = \lfloor \frac{1+1+0}{2} \rfloor = \lfloor 1 \rfloor = 1,\ s_1 = 0, c = 1$$

$$j = 2 : d = 1,\ s_2 = 0, c = 1$$

$$j = 3 : d = 1,\ s_3 = 1, c = 1$$

$$s_4 = 1$$

$\Longrightarrow$ 10011, so the sum is $(11001)_2$

Example 2.5. Find the product of $a = (110)_2$ and $b = (101)_2$

Solution.

$$
\begin{array}{r}
1\ 1\ 0 \\
\times\ 1\ 0\ 1 \\
\hline
1\ 1\ 0 \\
0\ 0\ 0\ \\
1\ 1\ 0\ \\
\hline
1\ 1\ 1\ 1\ 0
\end{array}
$$

Thus, $(110)_2.(101)_2 = (11110)_2$

ALGORITHM 3 Multiplication of Integers.

procedure *multiply*(a, b: positive integers)
{the binary expansions of a and b are $(a_{n-1}a_{n-2}\ldots a_1a_0)_2$
  and $(b_{n-1}b_{n-2}\ldots b_1b_0)_2$, respectively}
for $j := 0$ **to** $n - 1$
  if $b_j = 1$ **then** $c_j := a$ shifted j places
  else $c_j := 0$
{$c_0, c_1, \ldots, c_{n-1}$ are the partial products}
$p := 0$
for $j := 0$ **to** $n - 1$
  $p := add(p, c_j)$
return p {p is the value of ab}

Solution of Example 2.5

$$j = 0 : b_0 = 1 \rightarrow c_0 = 110$$
$$j = 1 : b_1 = 0 \rightarrow c_1 = 000\textcolor{red}{0}$$
$$j = 2 : b_2 = 1 \rightarrow c_2 = 110\textcolor{red}{00}$$

$$p = 0$$
$$j = 0 : p = 0 + 110 = 110$$
$$j = 1 : p = 110 + 0000 = 0110$$
$$j = 2 : p = 0110 + 11000 = 11110$$

$\rightarrow p = 11110$. Hence, the product $(11110)_2$

ALGORITHM FOR div AND mod

ALGORITHM 4 Computing div and mod.

procedure *division algorithm*(a: integer, d: positive integer)
$q := 0$
$r := |a|$
while $r \geq d$
  $r := r - d$
  $q := q + 1$
if $a < 0$ and $r > 0$ **then**
  $r := d - r$
  $q := -(q + 1)$
return (q, r) $\{q = a$ **div** d is the quotient, $r = a$ **mod** d is the remainder$\}$

Example 2.6. Find the quotient and the remainder in the division of 101 by 11.

Solution of Example 2.6

$q = 0, r = |101| = 101$

$$r = 101 \geq 11 = d \rightarrow r = 101 - 11 = 90, q = 0 + 1 = 1$$
$$r = 90 \geq 11 = d \rightarrow r = 79, q = 2$$
$$r = 79 \geq 11 = d \rightarrow r = 61, q = 3$$
$$r = 61 \geq 11 = d \rightarrow r = 57, q = 4$$
$$r = 57 \geq 11 = d \rightarrow r = 46, q = 5$$
$$r = 46 \geq 11 = d \rightarrow r = 35, q = 6$$
$$r = 35 \geq 11 = d \rightarrow r = 24, q = 7$$
$$r = 24 \geq 11 = d \rightarrow r = 13, q = 8$$
$$r = 13 \geq 11 = d \rightarrow r = 2, q = 9$$
$$r = 2 \geq 11 = d(!)$$

Hence, $(q = 9, r = 2)$

Modular Exponentiation

ALGORITHM 5 Fast Modular Exponentiation.

procedure *modular exponentiation*(b: integer, $n = (a_{k-1}a_{k-2}\ldots a_1a_0)_2$,
    m: positive integers)
$x := 1$
$power := b$ **mod** m
for $i := 0$ **to** $k - 1$
    if $a_i = 1$ **then** $x := (x \cdot power)$ **mod** m
    $power := (power \cdot power)$ **mod** m
return $x\{x$ equals b^n **mod** $m\}$

Example 2.7. Find 3^{644} mod 645. (We have $b = 3, m = 645, 644 = (1010000100)_2$).

Solution of Example 2.7

$x = 1$, $power = 3 \bmod 645 = 3$

$i = 0 : a_0 = 0 \to x = 1, power = 3^2 \bmod 645 = 9$

$i = 1 : a_1 = 0 \to x = 1, power = 9^2 \bmod 645 = 81$

$i = 2 : a_2 = 1 \to x = 1.81 \bmod 645 = 81, power = 81^2 \bmod 645 = 111$

$i = 3 : a_3 = 0 \to x = 81, power = 111^2 \bmod 645 = 66$

$i = 4 : a_4 = 0 \to x = 81, power = 66^2 \bmod 645 = 486$

$i = 5 : a_5 = 0 \to x = 81, power = 486^2 \bmod 645 = 126$

$i = 6 : a_6 = 0 \to x = 81, power = 126^2 \bmod 645 = 396$

$i = 7 : a_7 = 1 \to x = (81.396) \bmod 645 = 471, power = 396^2 \bmod 645 = 81$

$i = 8 : a_8 = 0 \to x = 471, power = 81^2 \bmod 645 = 111$

$i = 9 : a_9 = 1 \to x = (471.111) \bmod 645 = 36, power = 111^2 \bmod 645 = 66$

$\Longrightarrow x = 3^{644} \bmod 645 = 36.$

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Definition 3.1

1. A positive integer p greater than 1 is called **prime** (số nguyên tố) if the only positive factors (ước số) are 1 and p.
2. A positive integer that is greater than 1 and is not prime is called **composite** (hợp số).

Remark 3.1.

1. The integer 1 is NOT prime, because it has only one positive factor.
2. An integer n is composite if and only if there exists an integer a such that $a \mid n$ and $1 < a < n$.

Example 3.1. The integer 7 is prime because its only positive factors are 1 and 7, whereas the integer 9 is composite because it is divisible by 3.

The fundamental theorem of arithmetic

Theorem 3.1

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size.

Example 3.2. The prime factorizations of 100, 999, and 1024 are given by

$$100 = 2^2.5^2$$

$$999 = 3^3.37$$

$$1024 = 2^{10}$$

Trial Division

Theorem 3.2

If n is a composite integer, then n has a prime divisor (ước nguyên tố) less than or equal to $\sqrt{n}$.

Example 3.3. Show that 101 is prime.

Solution.
The only primes not exceeding $\sqrt{101} \approx 10.05$ are 2, 3, 5, and 7. Because 101 is not divisible by 2, 3, 5, or 7, it follows that 101 is prime.

Example 3.4. Find the prime factorization of 7007.

Solution. (Casio: $7007 \rightarrow = \rightarrow SHIFT \rightarrow FACT$)
$2 \nmid 7007$, $3 \nmid 7007$, $5 \nmid 7007$, $7 \mid 7007$
$7007/7 = 1001$, $1001/7 = 143$, $7 \nmid 143, 143/11 = 13$, 13 is a prime $\rightarrow$ completed.
$\Longrightarrow 7007 = 7.1001 = 7.7.143 = 7.7.11.13 = 7^2.11.13$

Theorem: THE INFINITUDE OF PRIMES

Theorem 3.3

There are infinite many primes.

Theorem 3.4: THE PRIME NUMBER THEOREM

The ratio of $\pi(x)$, the number of primes not exceeding x and $x/\ln x$ approaches 1 and grows with bound.

TABLE 2 Approximating $\pi(x)$ by $x/\ln x$.

x	$\pi(x)$	$x/\ln x$	$\pi(x)/(x/\ln x)$
10^3	168	144.8	1.161
10^4	1229	1085.7	1.132
10^5	9592	8685.9	1.104
10^6	78,498	72,382.4	1.084
10^7	664,579	620,420.7	1.071
10^8	5,761,455	5,428,681.0	1.061
10^9	50,847,534	48,254,942.4	1.054
10^{10}	455,052,512	434,294,481.9	1.048

Conjectures and Open Problems About Primes

1. **Goldbach's Conjecture:** Every even integer n, $n > 2$, is the sum of two primes. For example, $4 = 2 + 2$, $6 = 3 + 3$, $8 = 5 + 3$, $10 = 7 + 3$, $12 = 7 + 5$, and so on.
2. **The Twin Prime Conjecture:** Twin primes are pairs of primes that differ by 2, such as 3 and 5, 5 and 7, 11 and 13, 17 and 19, and 4967 and 4969.
3. **Green-Tao theorem:** There are arbitrarily long arithmetic progressions of primes. For example, Arithmetic progression of 3 primes: 3, 5, 7 (with common difference d=2); Arithmetic progression of 5 primes: 5, 11, 17, 23, 29 (with common difference d=6).

Greatest Common Divisors

Definition 3.2

Let a, b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called **the greatest common divisor** (ước chung lớn nhất) of a and b.
Notation: $\gcd(a, b)$

1. Method 1: Prime Factorization
   1.1 Factorize a and b into their prime factors:
   $$a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \; b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}$$
   1.2 Take the product of common prime factors with the smallest exponents:
   $$\gcd(a, b) = p_1^{\min(\alpha_1, \beta_1)} p_2^{\min(\alpha_2, \beta_2)} \cdots p_k^{\min(\alpha_k, \beta_k)}$$
2. Method 2: The Euclidean Algorithm.
3. ...

Example 3.5. $gcd(24, 36) \to gcd(24, 36) = 12$

Least Common Multiple

Definition 3.3

The Least common multiple (bội chung nhỏ nhất) of the positive integer a and b is the smallest integer that is divisible by both a and b.
Notation: $\operatorname{lcm}(a, b)$

1. Method 1: Prime Factorization
   1.1 Factorize a and b into their prime factors:
   $$a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}, \ b = p_1^{\beta_1} p_2^{\beta_2} \cdots p_k^{\beta_k}$$
   1.2 Take the product of common prime factors with the smallest exponents:
   $$\operatorname{lcm}(a, b) = p_1^{\max(\alpha_1,\beta_1)} p_2^{\max(\alpha_2,\beta_2)} \cdots p_k^{\max(\alpha_k,\beta_k)}$$
2. Method 2: The Euclidean Algorithm.
3. ...

Example 3.6. $lcm(12, 36) \to lcm(12, 36) = 36$

Definition 3.4

The integers a, b are **relatively prime** (nguyên tố cùng nhau) if $\gcd(a,b)=1$.

Definition 3.5

The integers $a_1, a_2, a_3, \ldots, a_n$ are **pairwise relatively prime** (đôi 1 nguyên tố cùng nhau) if $\gcd(a_i, a_j) = 1$ whenever $1 \leq i < j \leq n$.

Theorem 3.5

Let a, b be positive integers then $ab = \gcd(a,b) \cdot \operatorname{lcm}(a,b)$.

Example 3.7. $\gcd(8,12) = 4, \operatorname{lcm}(8,12) = 24 \to 8.12 = 4.24$

Example 3.8. Determine whether the integers 10, 17, and 21 are pairwise relatively prime and whether the integers 10, 19, and 24 are pairwise relatively prime.

The Euclidean Algorithm

Lemma 3.1

Let $a = bq + r$, where a, b, q, and r are integers. Then $\gcd(a, b) = \gcd(b, r)$.

ALGORITHM 1 The Euclidean Algorithm.

procedure *gcd*(a, b: positive integers)
$x := a$
$y := b$
while $y \neq 0$
    $r := x$ **mod** y
    $x := y$
    $y := r$
return x {gcd(a, b) is x}

Example 3.9. Find $\gcd(24, 36)$.

Solution of Example 3.9

$x = 24$
$y = 36$

$y = 36 \neq 0 : r = 24 \bmod 36 = 24, x = 36, y = 24$

$y = 24 \neq 0 : r = 36 \bmod 24 = 12, x = 24, y = 12$

$y = 12 \neq 0 : r = 24 \bmod 12 = 0, x = 12, y = 0$

$y = 0 \neq 0(!)$

$\rightarrow$ return 12. Hence, $gcd(24, 36) = 12$

Theorem 3.6: BEZOUT'S THEOREM

If a and b are positive integers, then there exist integers s and t such that $\gcd(a, b) = sa + tb$.

Lemma 3.2

If a, b, and c are positive integers such that $\gcd(a, b) = 1$ and $a \mid bc$, then $a \mid c$.

Theorem 3.7

Let m be a positive integer and let a, b, and c be integers. If $ac \equiv bc \pmod{m}$ and $\gcd(c, m) = 1$, then $a \equiv b \pmod{m}$.

Euler φ-function

Definition 3.6

$$\varphi : \mathbb{N}^* \longrightarrow \mathbb{N}^*$$

$$n \longmapsto \varphi(n) = \begin{cases} n\left(1-\dfrac{1}{p_1}\right)\left(1-\dfrac{1}{p_2}\right)\cdots\left(1-\dfrac{1}{p_k}\right), & n = p_1^{\alpha_1} p_2^{\alpha_2}\cdots p_k^{\alpha_k} \\ n-1, & n \text{ is prime} \end{cases}$$

$\varphi(n) =$ the number of positive integers $\leq n$ that are **relatively prime to n**.

Example 3.10. $\varphi(6) = 6\left(1-\dfrac{1}{2}\right)\left(1-\dfrac{1}{3}\right) = 2$ (1 and 5).

Example 3.11. Find $\varphi(10)$ and $\varphi(100)$

Quiz

Q1. Which pair of integers are relatively prime?

a. $(17, 51)$

b. $(5, 24)$

c. $(11, 121)$

d. None of the others

Q2. If a, b are positive integers such that $\gcd(a, b) = 5$ and $ab = 120$. Find $\operatorname{lcm}(a, b)$.

a. 24

b. 600

c. 120

d. None of the others

Q&A

Thank you for listening!